

SECTION 'B' (Short-Answer Questions) (15)

NOTE: Attempt any THREE part questions. All questions carries equal marks.

2. i) Draw a Pie diagram for the following table:

Items	Expenditure
Food	2300
Clothing	1000
House Rent	1200
Education	900
Miscellaneous	600

2. ii) The average wage of 4 men is Rs. 17 per hour and the average wage of further 6 men is Rs. 22 per hour. What is the average wage of the 10 men?

2. iii) If a fair coin is tossed three times:

a) What is probability of getting 3 Heads?

b) What is the probability of getting 2 Heads?

2. iv) The price of a Commodity for 6 years is given in the following table .

Year	2015	2016	2017	2018	2019	2020
Price	20	21	23	28	30	35

Calculate index number a) Taking 2015 as base year b) Using Chain base method.

2. v) Show that the sum of the deviations from their mean is Zero: 1.3, 7.0, 3.6, 4.1, 5.0.

SECTION 'C' (Detailed Answer Questions)(10)

NOTE: Attempt any ONE part questions. All questions carries equal marks.

3. Find Mean, Median and Mode of the following frequency distribution and comment on the results:

Daily Wages	No. of Employees
4-6	2
6-8	5
8-10	10
10-12	8
12-14	3
14-16	1

4. From the following data using 2019 as base year, find:

i) Laspyere's Index Number

ii) Paasche's Index Number

iii) Fisher Ideal Index Number

Commodities	Price		Quantity	
	2019	2020	2019	2020
A	64	75	270	290
B	58	68	185	200
C	18	21	130	137
D	50	60	134	145

نوٹ: سال 2020 میں گیارہویں اور بارہویں جماعت کے امتحانات نہیں ہوئے تھے۔

STATISTIC 2021

(AS PER CONDENSED SYLLABUS)

TIME: 2 Hours

(50 Marks)

NOTE:

- i) This section consists of 25 part questions and all are to be answered. Each question carries ONE marks.
- ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.
- iii) The use of calculator is allowed.

SECTION 'A' (Multiple Choice Questions)(25)

1. Choose the correct answer for each from the given options:

(i) A die is rolled once, the total sample points will be:

- * 6✓ * 12 * 36 * 2

(ii) Range is calculated as:

- * Max. value - Min. value ✓ * Min. value - Max. value
* Mean - Mode * Median - Mean

(iii) The color of eyes represents:

- * Quantitative data * Qualitative data ✓
* Discrete data * Continuous data

(iv) In tossing two coins together the total possible outcomes are:

- * 2 * 4✓ * 6 * 16

(v) Mean of the data 1, 4, 2, 3, 5 is:

- * 5 * 10 * 15 * 3✓

(vi) 5P_2 is equal to:

- * 10 * 20✓ * 120 * 70

(vii) First hand information is called:

- * Mean * Mode
* Primary data ✓ * Secondary Data

(viii) Circular diagram of data is called:

- * Pie-diagram ✓ * Multiple bar diagram
* Ogive * Bar diagram

(ix) The part of the population under study is called:

- * Parameter * Sample ✓ * Mean * Statistic

(x) $0!$ is equal to:

- * 0 * Infinity * 1✓ * 10

(xi) The median of the data 1, -1, 0, 2, -2 is:

- * 6 * 0✓ * -2 * +2

(xii) In a symmetrical distribution Mean, Median and Mode are:

- * Equal ✓ * Greater than Zero
* Less than Zero * Equal to Zero

(xiii) The information collected from magazine is:

- * Primary data * Continuous data ✓
* Secondary data ✓ * Discrete data

(xiv) 5 people out of 7 can be selected in:

- * 10 ways * 5 ways * 50 ways * 21 ways ✓

(xv) The graph of frequency distribution is:

- * Bar diagram * Frequency curve
* Pie-diagram * Ogive

(xvi) The probability always lies between:

- * 5 to 10 * 0 to 1 ✓ * -1 to +1 * 1 to 10

(xvii) Fisher Index Number is given by:

- * $\sqrt{L \times P}$ ✓ * $\sqrt{L - P}$ * $\sqrt{L + P}$ * $\sqrt{L/P}$

(xviii) A characteristic of population is called:

- * Mean * Mode * Median * Parameter ✓

(xix) Statistics deals with:

- * Theory * Graph * Data ✓ * Mean

(xx) If $P(A) = 0.6$, $P(B) = 0.8$, $P(A \cap B) = 0.84$ then $P(A \cup B)$ equal to :

- * -0.25 * 0.48 * 0.56 ✓ * 1.40

(xxi) The most repeated value in data is called:

- * Mode ✓ * Mean * Median * Constant

(xxii) Median can be located by:

- * Bar diagram * Histogram * Pie-Diagram * Ogive ✓

(xxiii) Mode can be located by:

- * Histogram ✓ * Bar diagram
* Ogive * Pie-diagram

(xxiv) If $\sum(x - A) = 0$, then Arithmetic Mean will be:

- * $\bar{x} = 20$ * $\bar{x} = A$ ✓ * $\bar{x} = 0$ * $\bar{x} = 5$

(xxv) The sum of relative frequencies must be equal to:

- * 3 * 2 * 1 ✓ * 5

STATISTICS

Time: 1:45 Hour (Regular & Private)

Marks:40

SECTION 'B' (SHORT-ANSWER QUESTIONS)(25)

Note: Attempt 5 part questions from this section.

2(i) Differentiate between the following:

- (a) Discrete and Continuous variable
- (b) Primary and Secondary data

(ii) Draw a Multiple Bar Diagram of the following data:

Country	Birth Rate	Death Rate
Pakistan	50	20
India	45	15
England	15	10
Egypt	40	15
Japan	10	14

(iii) In how many ways can the letters of the following words be arranged? (a) TOMORROW (b) YESTERDAY

(iv) The following table shows the prices of an item for 5 years. Calculate the index number for all years, taking an average of 1990 and 1991 as base year.

Year	1990	1991	1992	1993	1994	1995	1996	1997
Price	45	50	60	65	70	72	68	83

(v) The probability that Jawwad will pass the test $\frac{2}{3}$ and that for Daniyal is $\frac{3}{4}$. Find the probability that

(a) both will pass the test (b) both will not pass the test

(vi) How many three digit numbers can be formed from the digits 1, 4, 5, 7, 8, 9 if the digits are:

- (a) repeated (b) not repeated

(vii) The mean of ten numbers is 8. If the eleventh number is included in the data, the mean becomes 9. What is the value of the eleventh number?

(viii) The following data refers to the prices of rice in a city:

125	126	135	132	130
128	123	122	131	120
130	135	121	128	126
132	134	135	137	132
138	129	130	124	126

Prepare a frequency table with class interval of 4 rupees each such that the lower class limit of first class is 119.

SECTION C (DETAILED-ANSWER QUESTION)(15)

Note: Attempt Two questions from this section

3. Calculate the price index numbers for the following data by applying:

Commodities	Price		Quantity	
	2008	2009	2008	2009
A	7	11	40	50
B	3	3	80	100
C	5	7	50	50

4. Find the mean, the median and the mode for the following frequency distribution:

Class Interval	2 - 4	5 - 7	8 - 10	11 - 13	14 - 16
Frequency	2	8	12	4	2

5. A bag contains 3 white balls, 4 black balls and 5 red balls. If 3 balls are drawn at random, find the probability that: (i) all the three balls are red (ii) Two balls are red and one ball is black. (iii) all three balls are different colour.

STATISTICS

2019

Time: 15 Minutes (Regular & Private) Marks: 10

SECTION "A" (MULTIPLE CHOICE QUESTIONS)

1. Choose the correct answer for each from the given options:

(i) If $\bar{x} = 10$ and $y = 3z + 2$, then y is:

- 30 • 32 • 2 • 10

(ii) Value index Number is equal to:

- $\frac{\sum p_o q_n}{\sum p_n q_o}$ • $\frac{\sum p_o q_o}{\sum p_n q_n}$ • $\frac{\sum p_n q_o}{\sum p_o q_o}$ • $\frac{\sum p_n q_n}{\sum p_o q_o}$

(iii) The limit of probability is:

- -1 to +1 • 0 to 1 • 1 to 2 • -2 to -1

(iv) If two dice are rolled once, then the probability of getting the same numbers is: • $\frac{1}{5}$ • $\frac{1}{2}$ • $\frac{1}{6}$ • 0

(v) The data obtained from the Federal Bureau of Statistics is called: • Secondary data

- Primary data • continuous data • complex data

(vi) The median of the data 4, 6, 3, 2, 0, 1, 7 is:

- 4 • 3 • 0 • 2

(vii) If the mean of the data of 25 values is 20, then $\sum x$ is =:

- 1.25 • 10 • 50 • 500

(viii) If $P(A) = \frac{1}{3}$, $P(B/A) = \frac{3}{10}$, then $P(A \cap B) =$:

- $\frac{9}{10}$ • $\frac{2}{10}$ • $\frac{6}{10}$ • $\frac{1}{10}$

(ix) The number of different arrangements of four persons to sit around a table is: • 4 • 6 • 16 • 26

(x) If $\sum (x - A) = 0$, then:

- $A < \bar{x}$ • $A > \bar{x}$ • $A = 0$ • $A = \bar{x}$

STATISTICS

2018

Time: 1 hours 45 Minutes

Max. Marks: 40

SECTION 'B' (SHORT-ANSWER QUESTIONS)(50)

NOTE: Attempt any 05 question from this section.

2(i) Differentiate between primary and secondary data, and give at least two source of each kind.

(ii) Complete the following table:

C.I.	Frequency	Relative frequency	Cumulative frequency
1 – 5	?	?	?
6 – 10	21	?	27
11 – 15	?	0.38	65
16 – 20	19	0.19	84
21 – 25	16	?	?

(iii) Draw histogram and frequency polygon for the following frequency distribution:

Class Interval	2 – 3	4 – 5	6 – 7	8 – 9	10 – 11
Frequency	1	3	5	4	2

(iv) For the data 101, 102, 103, 104, 105 and $y = x - 100$, find $\bar{x}$ and $\bar{y}$:

(v) The mean height of 45 student of a class is 60" and the mean height of 55 student of other class is 57". Find the combined mean.

(vi) Two fair dice are thrown together. Find the probability that the sum of points on the top of both dice is 9 or 10.

(vii) In how many ways can the world UNIVERSITY be arranged?

(viii) Let A and B are two events in sample space such that $P(A) = \frac{3}{4}$, $P(B) = \frac{1}{4}$ and $P(A \cup B) = \frac{5}{6}$ Find $P(A \cap B)$

SECTION 'C' (DETAILED- ANSWER QUESTIONS)

NOTE: Attempt any Two questions from this section

3. The following table is about mark awarded to 30 student

23	32	36	41	10	25	41	43	15	34
17	05	01	16	08	15	46	45	39	33
12	32	18	31	24	28	33	27	34	04

(a) Construct frequency distribution table, making classes 1 – 5, 6 – 10, 11 – 15, and so on. (b) Make cumulative frequency and relative frequency for the above data.

4. Calculate the median, the mode and locate both graphically for the following data.

Age (years)	08–12	12–16	16–20	20–24	24–28	Total
No. of Policy Holders	10	18	32	17	07	84

5. Calculate Laspeyre's, Paasche's and Fisher's index numbers by taking 1990 as base year.

Commodity	1990		1995	
	Price	Quantity	Price	Quantity
A	22	28	24	26
B	25	30	26	25
C	24	34	25	30
D	20	39	22	33

STATISTICS

2018

Time: 20 Minutes

Max. Marks: 20

SECTION "A" (MULTIPLE CHOICE QUESTION)

1. Choose the correct answer for each from the given options:

- (i) In Passche's weighted index number, weight is:
♦ Base year price ♦ Base year quantity
♦ Current year price ♦ Current year quantity
- (ii) Two events, which cannot occur together, are called:
♦ Joint events ♦ Composite event
♦ Mutually exclusive events
♦ Not mutually exclusive events
- (iii) If $P(A) = 0.4$, then $P(\bar{A}) =$:
♦ 0.4 ♦ 0.2 ♦ 0.6 ♦ 0.1
- (iv) For the data 5, 8, 10, 6, 10, 12, 15, 10, 6, mode is:
♦ 6 ♦ 15 ♦ 10 ♦ 20
- (v) If two fair coins are tossed, then the probability of getting one tail is:
♦ 0.25 ♦ 0.50 ♦ 0.70 ♦ 1
- (vi) The arithmetic mean of four values is 25, then is $\sum x$ is :
♦ 25 ♦ 50 ♦ 80 ♦ 100
- (vii) The characteristic of population is called:
♦ average ♦ statistic ♦ parameter ♦ variable
- (viii) The sum of deviations measured from its mean is always:
♦ minimum ♦ zero ♦ one ♦ maximum
- (ix) The number of arrangements of the letters of the word TYPE is: ♦ 4 ♦ 6 ♦ 24 ♦ 30
- (x) The graph of cumulative frequency distribution is called:
♦ Ogive ♦ Histogram ♦ Polygon ♦ Simple Bar diagram